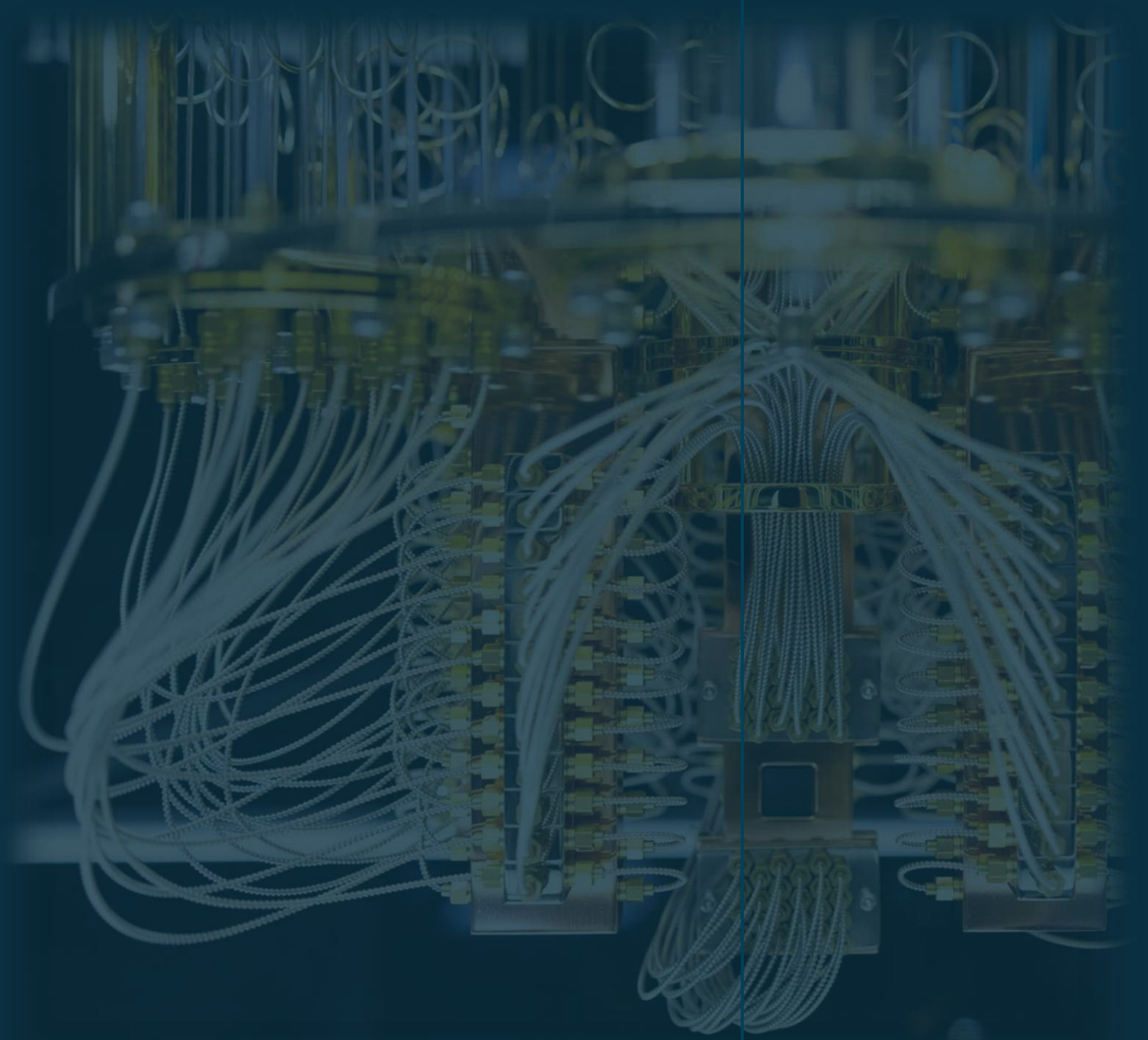
Quantum Security

Preparing for a Post-Quantum World

Nick Nieuwenhuis
Security Architect & Researcher

May 2, 2025

BSIDES



Please find the full recording on Youtube

“I think I can safely say that nobody understands quantum mechanics.”

Richard Feynman

Introduction



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LECTORAAT CYBER SECURITY

Microsoft CERTIFIED
CYBERSECURITY ARCHITECT
EXPERT

What to expect from this session?

- A (brief) history of quantum physics
- How quantum computers work and what concepts they use
- Opportunities & risks involved with quantum computing
- Cyber security in a post-quantum world

Introduction to Quantum Physics

Particles

“In a hole in the ground there lived a hobbit.”

J.R.R. Tolkien – The Hobbit

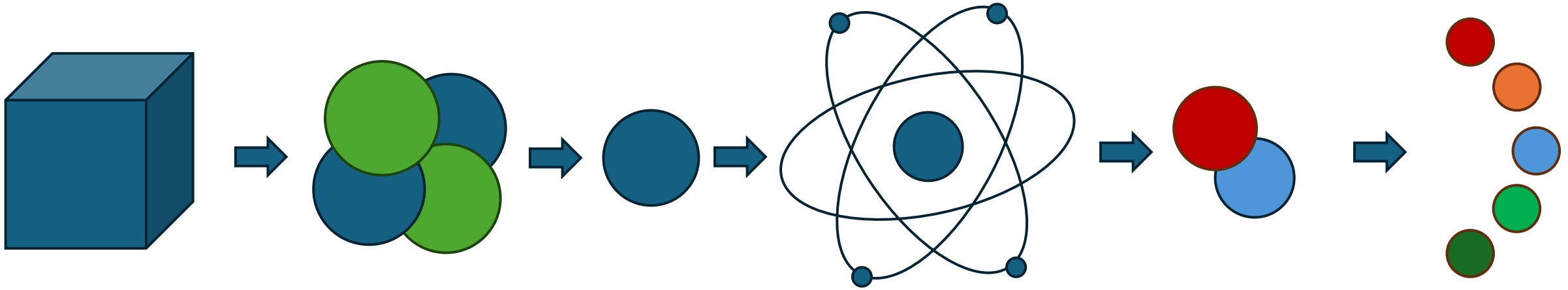
Particles

“a a b d e e e e g h h h i i i i l l n n o o r r t t v”

Particles

“In a hole in the ground there lived a hobbit.”

Particle Hierarchy



Matter

Molecule

Atom

Nucleus with electrons

Protons Neutrons

Elementary Particles

From Newtonian Physics to Quantum Physics | A Timeline



Isaac Newton

1687

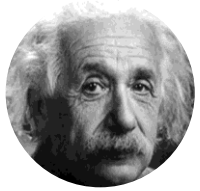
Defined the laws of Motion and Gravity



Max Planck

1900

“Energy comes in discrete packets (Quanta)**”**



Albert Einstein

1905

Photoelectric effect & Relativity



Niels Bohr

1913

Quantized model of the Atom



Erwin Schrödinger

1926

Wave Mechanics & Schrödinger equation

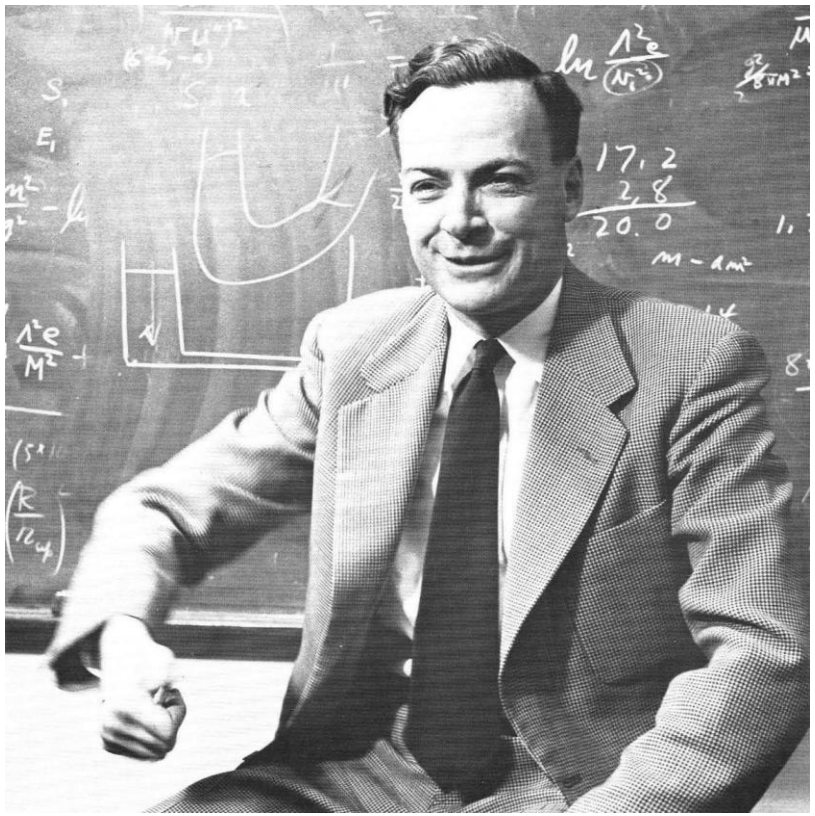


Werner Heisenberg

1927

First quantum theory and uncertainty principle

Richard Feynman



Simulating Physics with Computers

Richard P. Feynman

Department of Physics, California Institute of Technology, Pasadena, California 91107

Received May 7, 1981

1. INTRODUCTION

On the program it says this is a keynote speech—and I don't know what a keynote speech is. I do not intend in any way to suggest what should be in this meeting as a keynote of the subjects or anything like that. I have my own things to say and to talk about and there's no implication that anybody needs to talk about the same thing or anything like it. So what I want to talk about is what Mike Dertouzos suggested that nobody would talk about. I want to talk about the problem of simulating physics with computers and I mean that in a specific way which I am going to explain. The reason for doing this is something that I learned about from Ed Fredkin, and my entire interest in the subject has been inspired by him. It has to do with learning something about the possibilities of computers, and also something about possibilities in physics. If we suppose that we know all the physical laws perfectly, of course we don't have to pay any attention to computers. It's interesting anyway to entertain oneself with the idea that we've got something to learn about physical laws; and if I take a relaxed view here (after all I'm here and not at home) I'll admit that we don't understand everything.

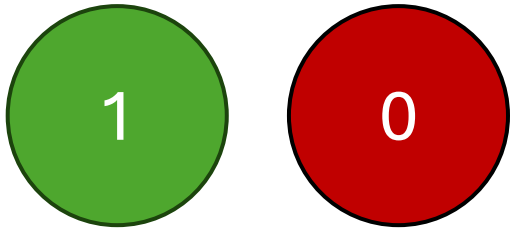
The first question is, What kind of computer are we going to use to simulate physics? Computer theory has been developed to a point where it realizes that it doesn't make any difference; when you get to a *universal computer*, it doesn't matter how it's manufactured, how it's actually made. Therefore my question is, Can physics be simulated by a universal computer? I would like to have the elements of this computer *locally interconnected*, and therefore sort of think about cellular automata as an example (but I don't want to force it). But I do want something involved with the

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"*Nature isn't classical... and if you want to simulate nature, you need a quantum computer*" - Richard Feynman

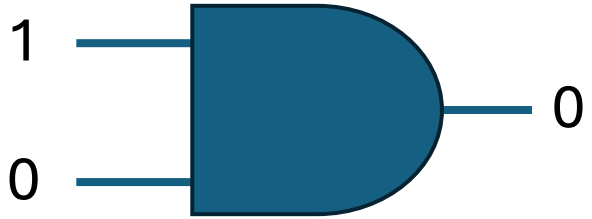
Key Concepts of Quantum Systems

The basics of classical Computers

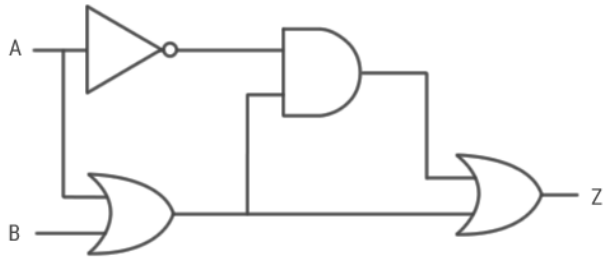


Binary Digits
‘Bits’

A unit of information



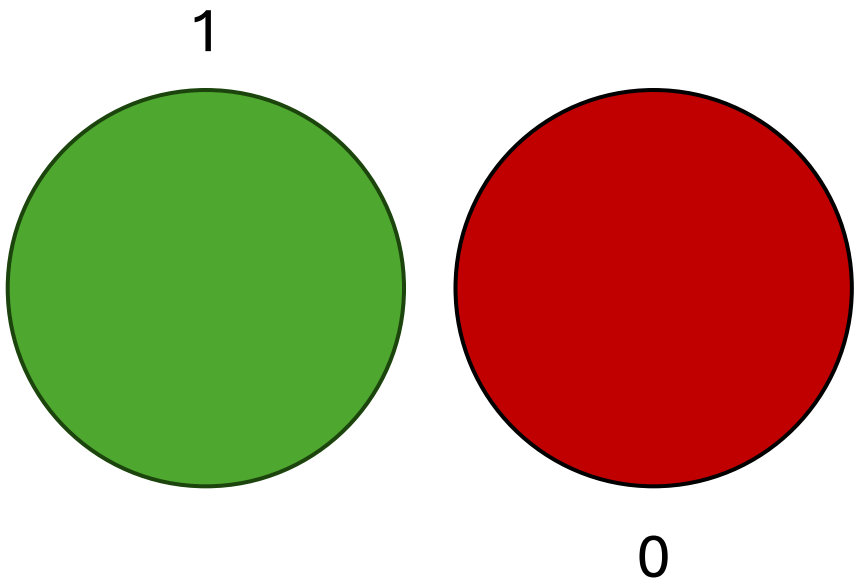
Gates



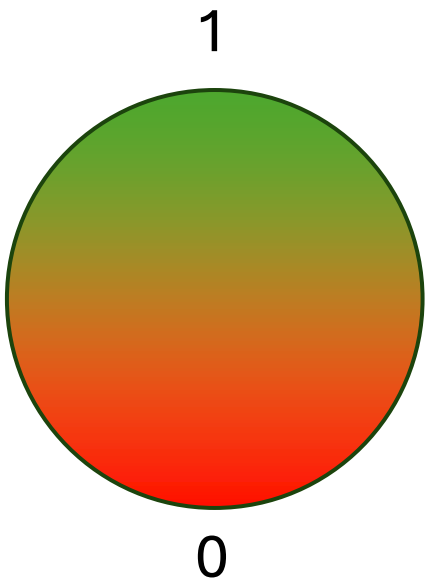
Circuits

A unit of computation

Superposition

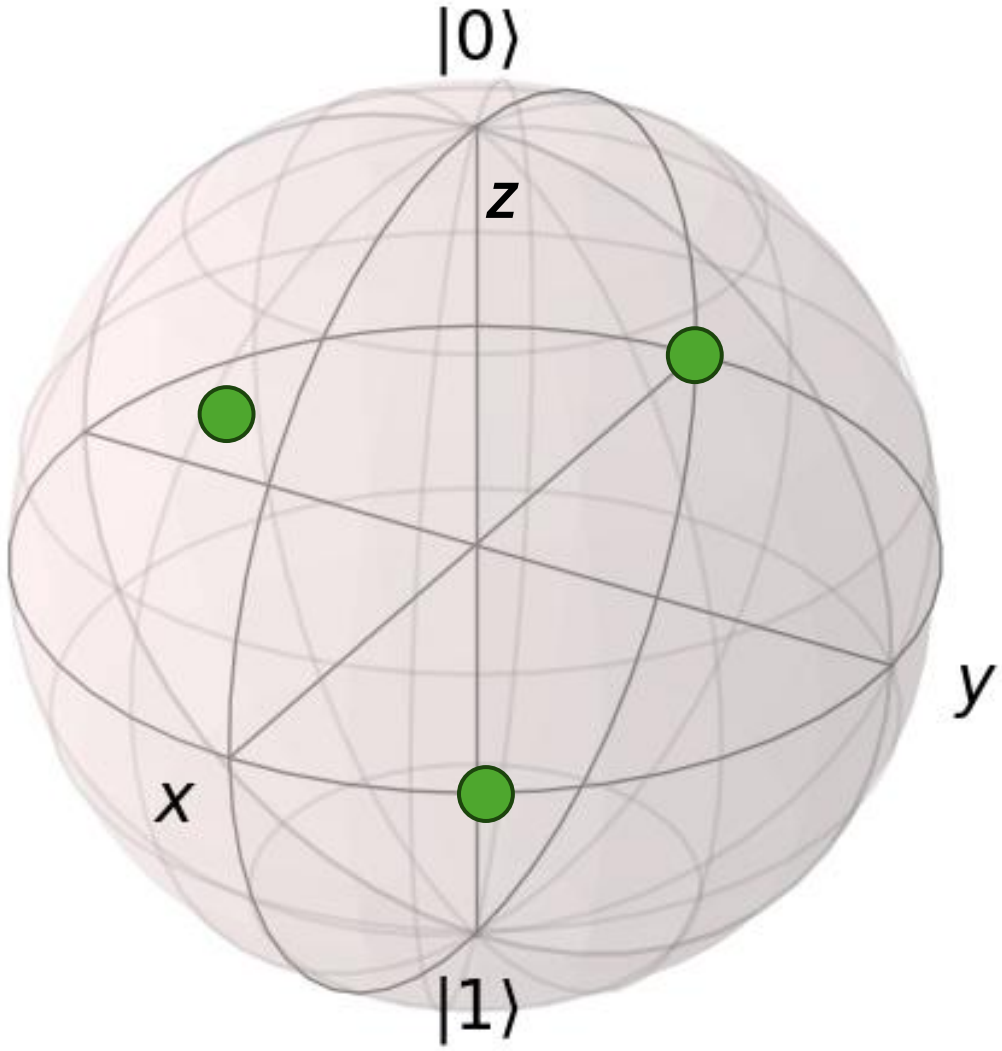


Binary Digits
‘Bits’



Quantum Bits
‘Qubits’

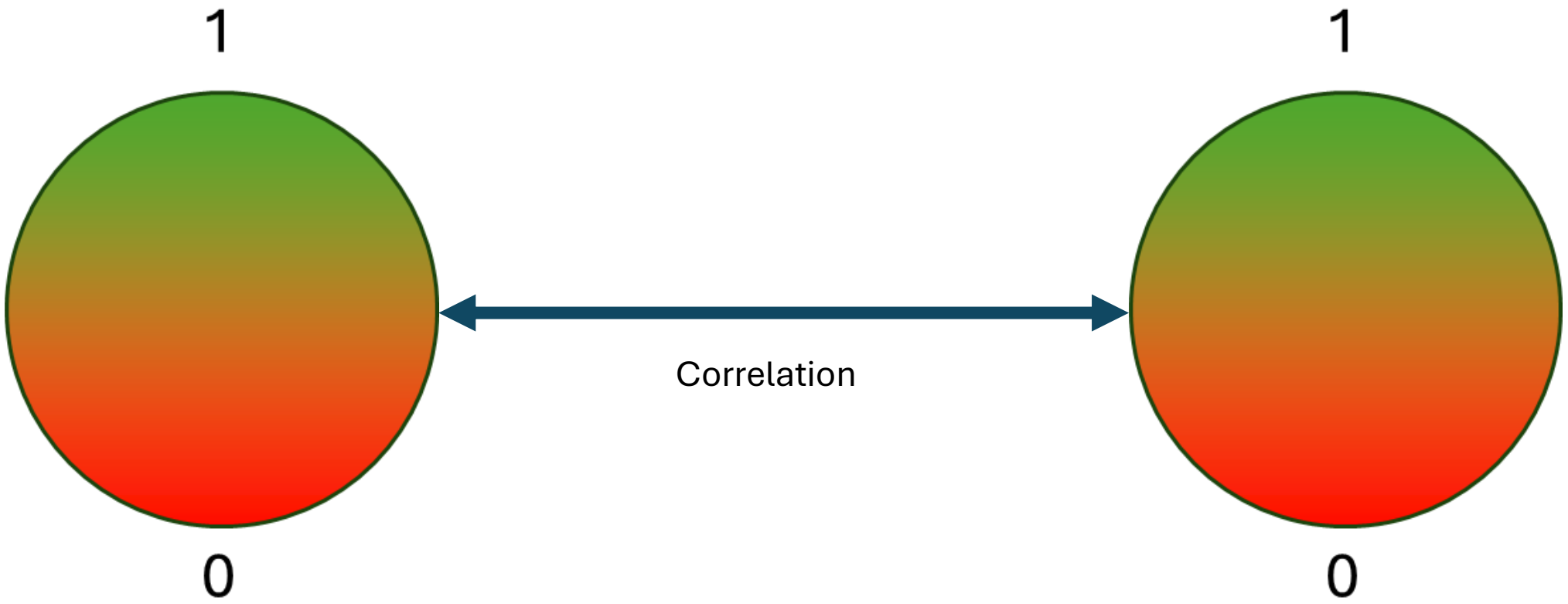
Superposition | Bloch Sphere



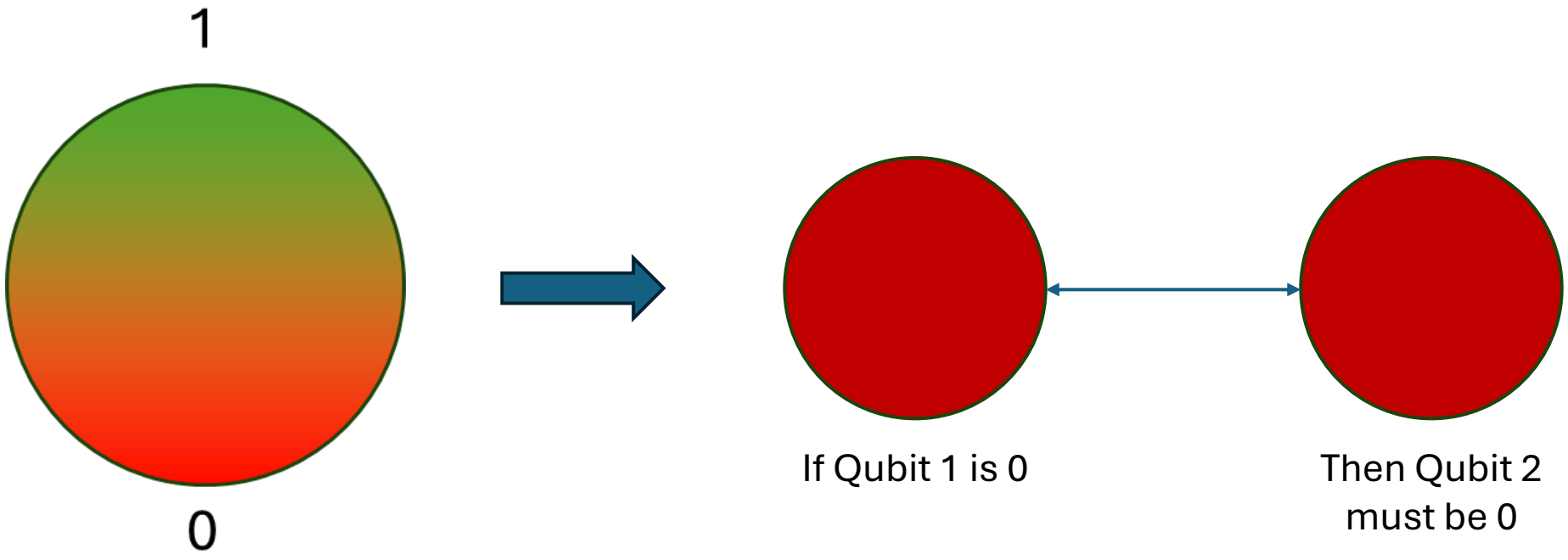
Measurement

Collapses the quantum state
into a classical state

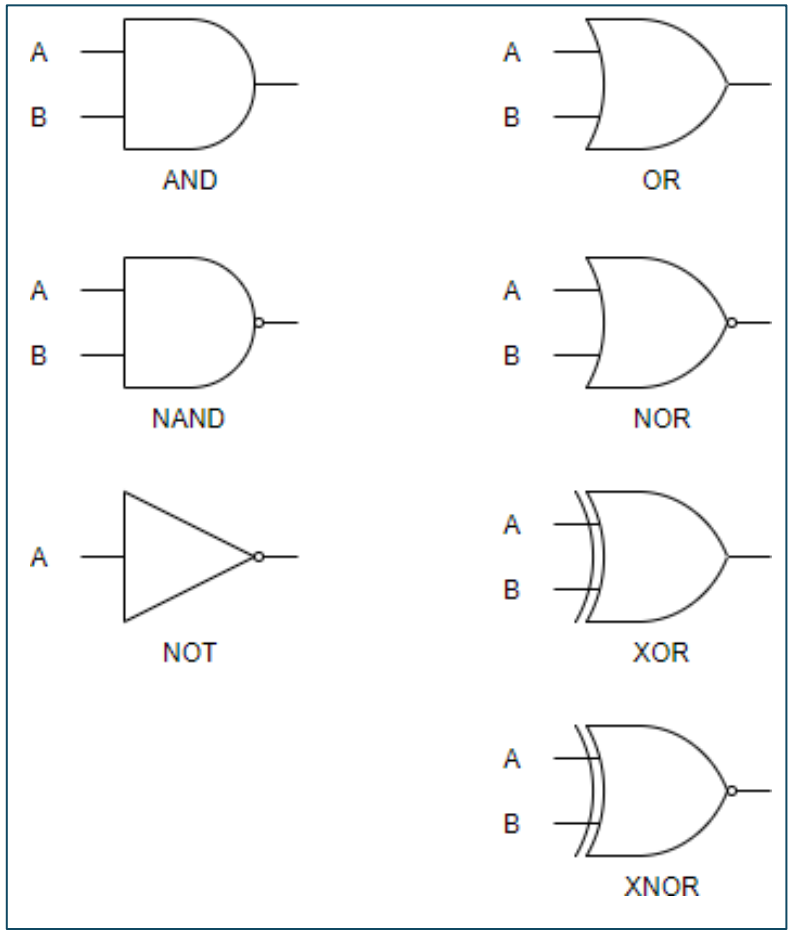
Entanglement | Correlation between two qubits



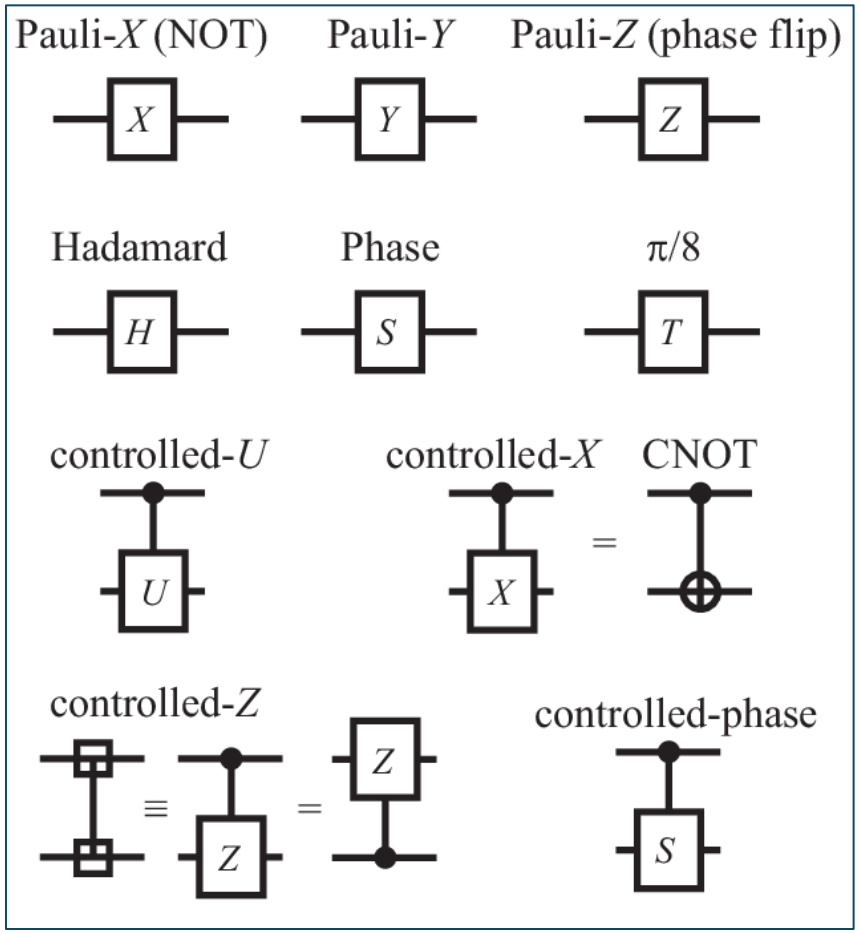
Entanglement | Spooky action at a distance



Quantum gates | Manipulating qubits into desired outcomes

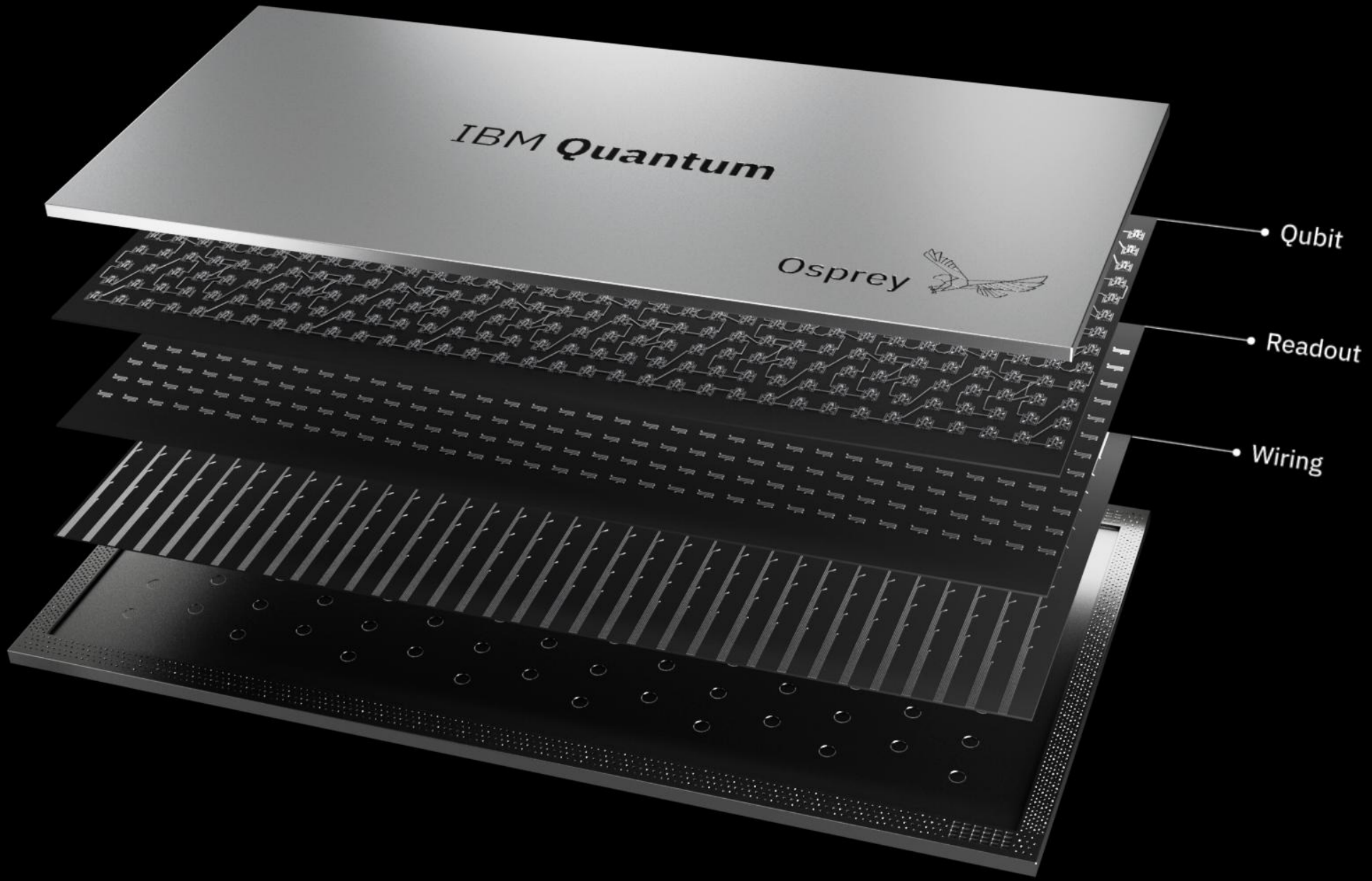


Classical gates



Quantum gates
(not exhaustive)




IBM Quantum
Osprey
Qubit
Readout
Wiring

Development Roadmap

IBM **Quantum**

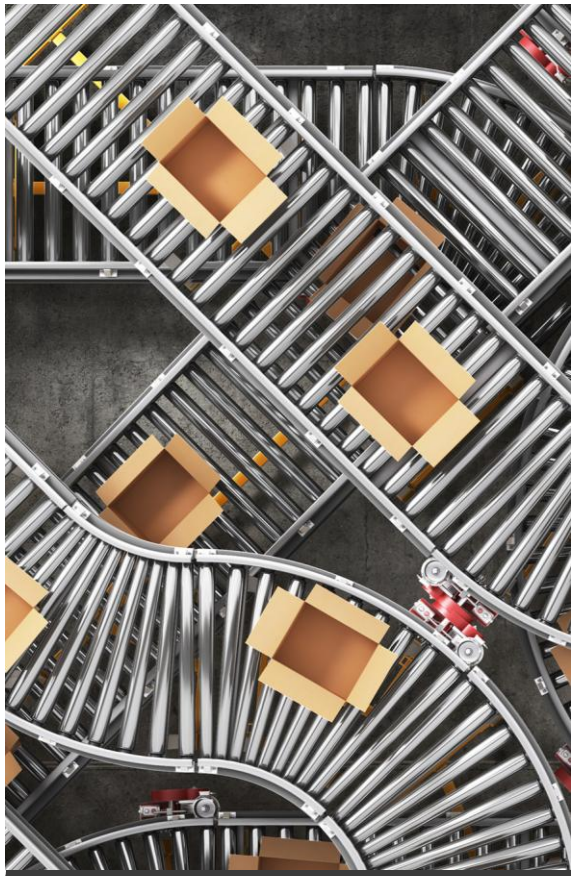
	2016–2019 ✓	2020 ✓	2021 ✓	2022 ✓	2023 ✓	2024	2025	2026	2027	2028	2029	2033+
	Ran quantum circuits on the IBM Quantum Platform	Released multi-dimensional roadmap publicly with initial aim focused on scaling	Enhanced quantum execution speed by 100x with Qiskit Runtime	Brought dynamic circuits to unlock more computations	Enhanced quantum execution speed by 5x with Quantum Serverless and execution modes	Improve quantum circuit quality and speed to allow 5K gates with parametric circuits	Enhance quantum execution speed and parallelization with partitioning and quantum modularity	Improve quantum circuit quality to allow 7.5K gates	Improve quantum circuit quality to allow 10K gates	Improve quantum circuit quality to allow 15K gates	Improve quantum circuit quality to allow 100M gates	Beyond 2033, quantum-centric supercomputers will include 1000's of logical qubits unlocking the full power of quantum computing
Data scientists						Platform						
						Code assistant	Functions	Mapping collections	Specific libraries			General purpose QC libraries
Researchers					Middleware							
					Quantum Serverless ✓	Transpiler service	Resource management	Circuit knitting x p	Intelligent orchestration			Circuit libraries
Quantum physicists			Qiskit Runtime									
	IBM Quantum Experience ✓		QASM 3 ✓	Dynamic circuits ✓	Execution modes ✓	Heron (5K)	Flamingo (5K)	Flamingo (7.5K)	Flamingo (10K)	Flamingo (15K)	Starling (100M)	Blue Jay (1B)
	Early ✓ Canary 5 qubits Albatross 16 qubits Penguin 20 qubits Prototype 53 qubits	**Falcon** ✓ Benchmarking 27 qubits		**Eagle** ✓ Benchmarking 127 qubits		Error mitigation 5k gates 133 qubits Classical modular Up to 133x3 = 399 qubits	Error mitigation 5k gates 156 qubits Quantum modular Up to 156x7 = 1092 qubits	Error mitigation 7.5k gates 156 qubits Quantum modular Up to 156x7 = 1092 qubits	Error mitigation 10k gates 156 qubits Quantum modular Up to 156x7 = 1092 qubits	Error mitigation 15k gates 156 qubits Quantum modular Up to 156x7 = 1092 qubits	Error correction 100M gates 200 qubits Error corrected modularity	Error correction 1B gates 2000 qubits Error corrected modularity

✓ Executed by IBM

On target

Applications of Quantum Computing

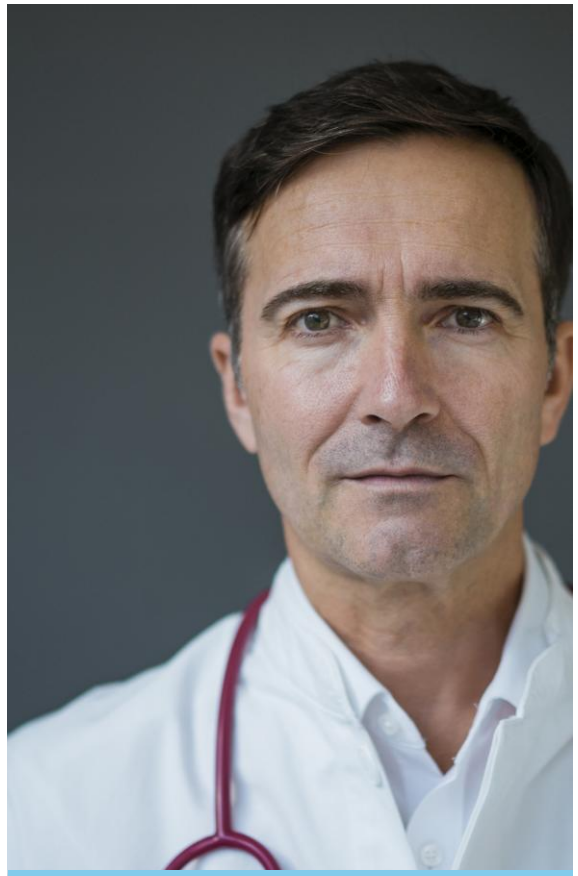
Applications of Quantum Computers



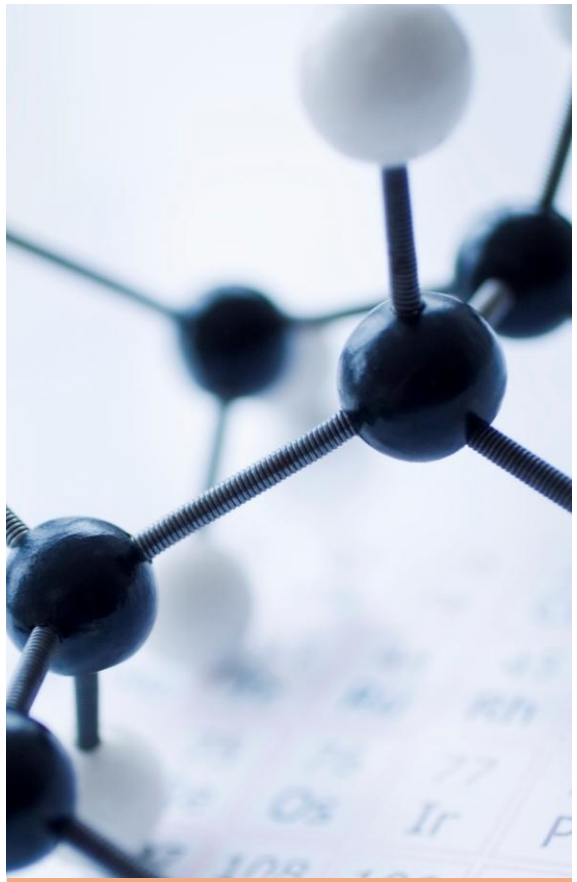
Logistics



Climate Change



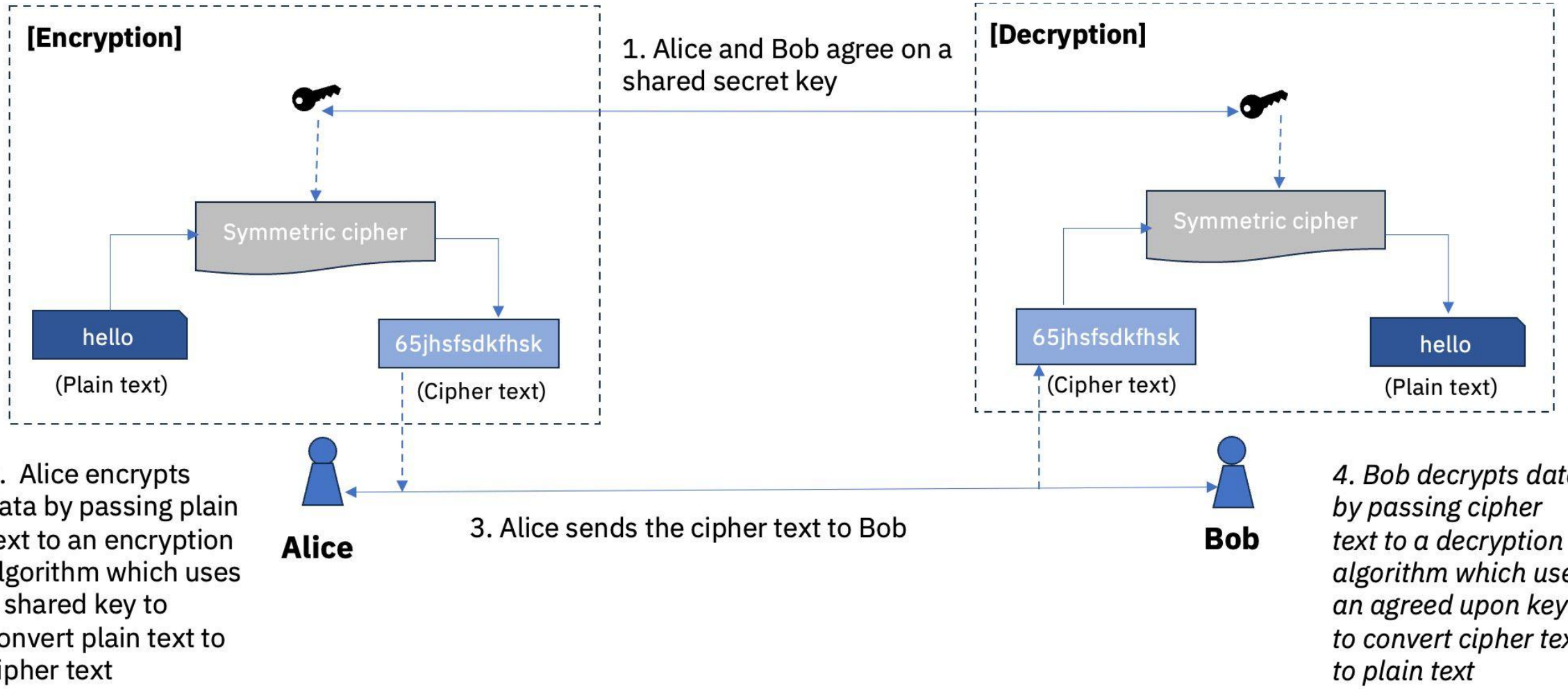
Healthcare



Chemistry

Let’s talk about cryptography

Symmetric Cryptography | Private Key Encryption



Symmetric cryptography | Common algorithms and key sizes

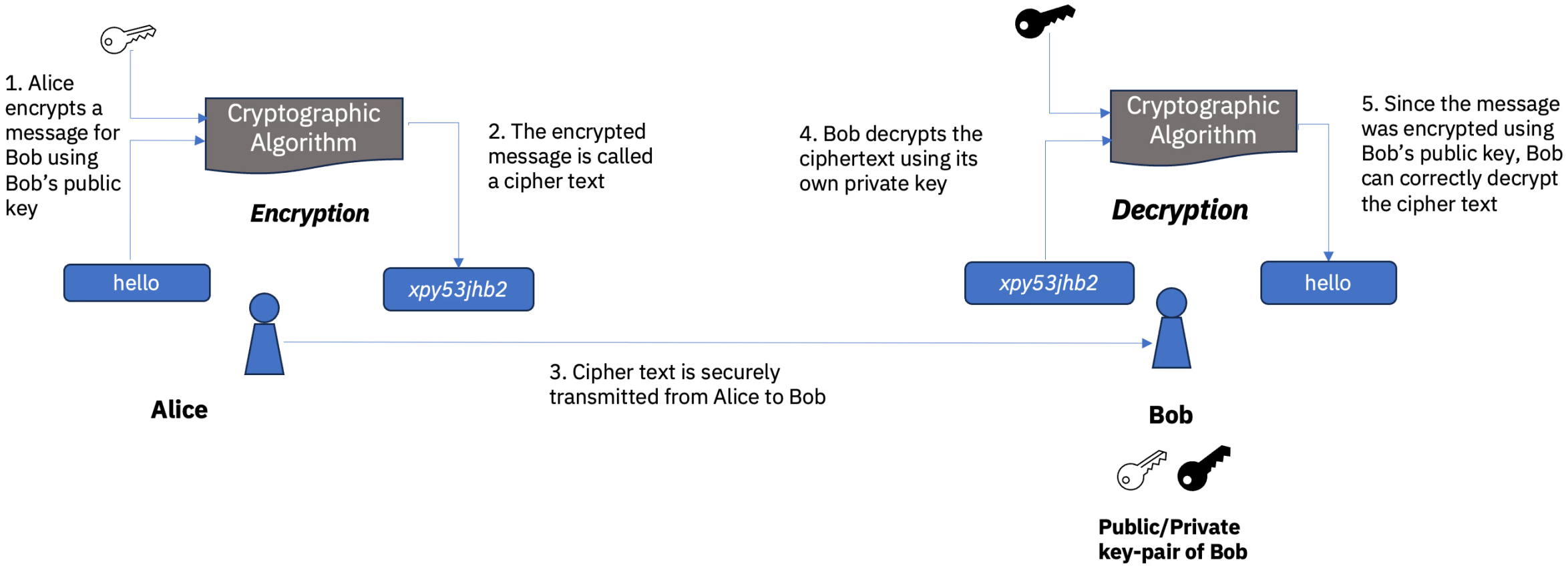
Algorithm	Key Sizes (bits)	Quantum-safe?
Data Encryption Standard (DES) 3DES	56 3x 56 (164)	No, disable where possible
Rivest Cipher 4 (RC4)	40 – 2048	No, already broken
Advanced Encryption Standard (AES)	128 192 **256**	Yes, but upgrade key size
Blowfish / Twofish	448 256	Yes

Symmetric cryptography | Quantum risks

1. Quantum brute-force attacks offer quadratic speed-up, using Grovers Algorithm

2. Doubling key-sizes mitigate this issue, but come at a cost:
   - Requires more compute power for encryption and decryption
   - Slower performance
   - More memory required
   - More energy used

Asymmetric Cryptography | Public Key Cryptography



Asymmetric cryptography | Common algorithms and key sizes

Algorithm	Key Sizes (bits)	Quantum-safe?
RSA (Rivest-Shamir-Adleman)	1024 (deprecated) 2048 3072 4096	No, broken by Shors' algorithm
Digital Signature Algorithm (DSA)	1024 (deprecated) 2048 3072	No, broken by Shors' algorithm
Diffie-Hellman (DH)	2048 3072 4096	No, broken by Shors' algorithm
Elliptic Curve Diffie Hellman (ECDH)	224, 256, 384, 521	No, broken by Shors' algorithm
Elliptic Curve Digital Signature Algorithm (ECDSA)	224, 256, 384, 521	No, broken by Shors' algorithm

Asymmetric cryptography | Quantum risks

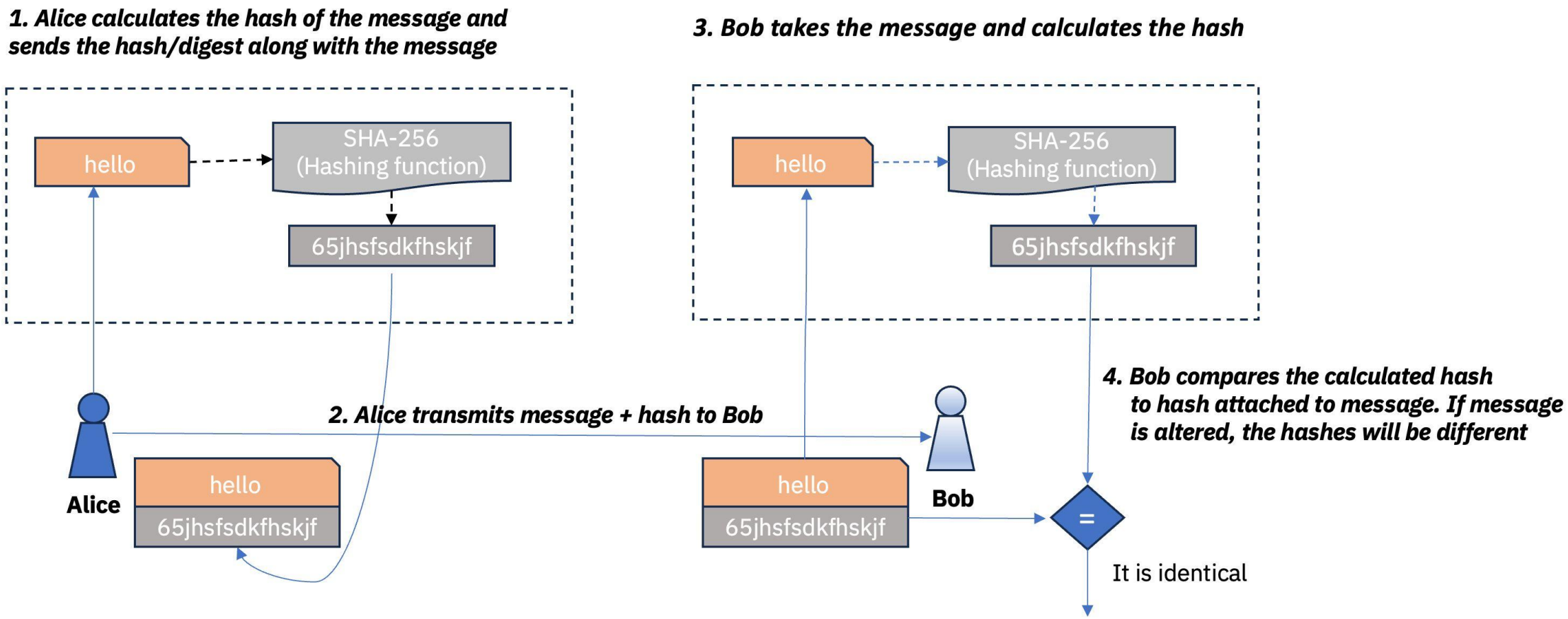
1. All the asymmetric cryptographic algorithms are susceptible to Shor's algorithm, a quantum algorithm that:

- Can factor large integers in polynomial time, breaking RSA.
- Can solve the Discrete Logarithm Problem in polynomial time, breaking both Diffie-Hellman (DH), DSA, and Elliptic Curve Cryptography (ECDSA/ECDH).

In normal words:

Quantum computers make it much easier to break the math behind modern public-key encryption!

Hash functions | Overview



Hash functions | Quantum risks

- Hashing algorithms (such as SHA-256) are susceptible to:
  - Pre-image attacks: Reverting a hash to its input
  - Collision attacks: Two different inputs leading to the same hash

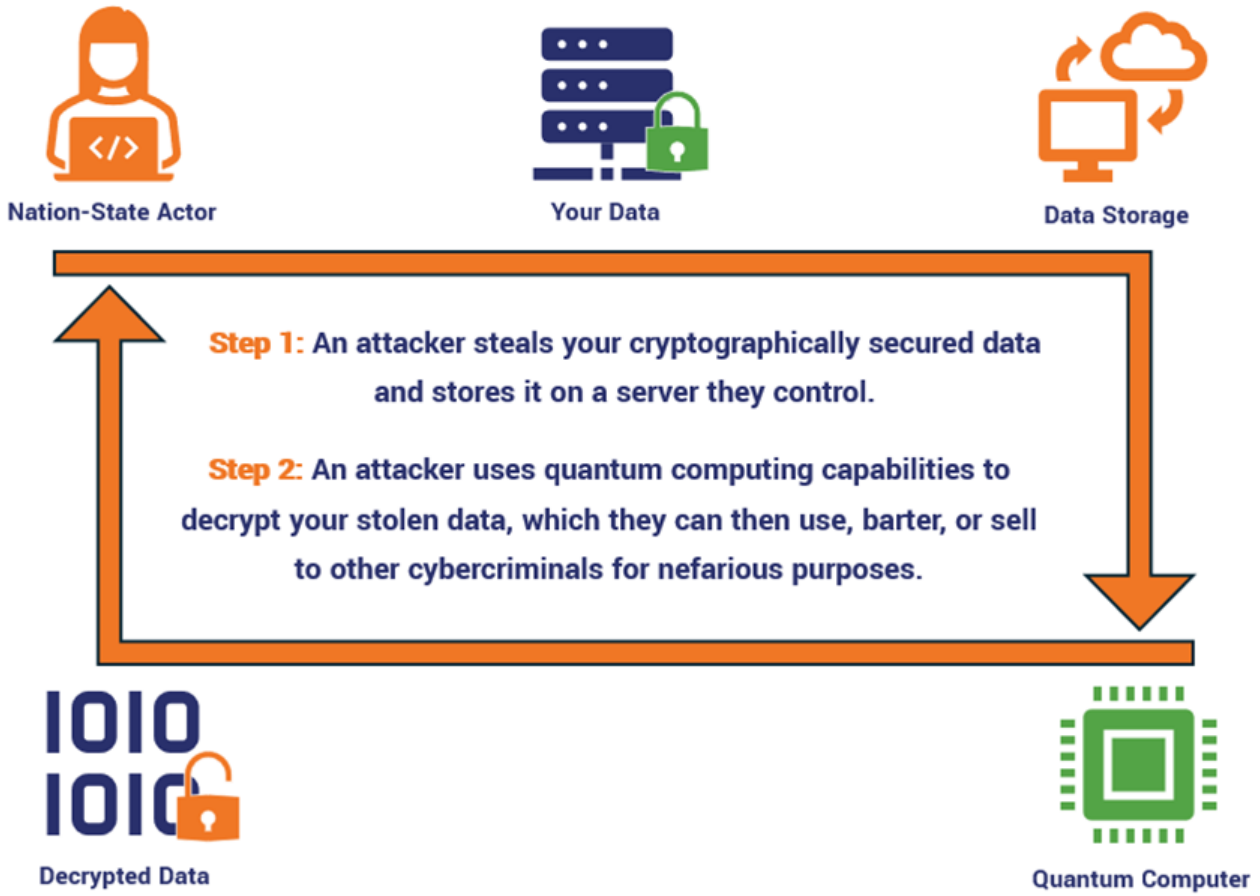
- The longer the hash, the harder it is to break:
  - Migrate away from SHA-128 to SHA-256 or SHA-512

Cybersecurity in a post-quantum world

Post-Quantum Cryptography (PQC) | Definition

“Post-Quantum Cryptography (PQC) encompasses efforts to **identify** and **develop** cryptographic schemes that can **withstand** attacks both from **quantum** and **classical** computing systems.

Harvest-Now & Decrypt Later | A current risk?



Post-Quantum Cryptography (PQC) | Finalists

NIST standardization efforts led to 4 finalists:

PCQ Algorithm	Replaces	Application
CRYSTALS-Kyber	RSA/Diffie-Hellman	Secure key exchange
CRYSTALS-Dilithium	RSA/ECDSA	Digital signatures
FALCON	ECDSA	Digital signatures
SPHINCS+	ECDSA	Digital signatures

It’s time to start building quantum-readiness!

Post-Quantum Cryptography (PQC) | Migration scenarios

Cryptographic scheme	Advice	When?
Symmetric cryptography (AES, Twofish)	Move to larger key size where applicable (AES-256)	Now
Asymmetric cryptography (RSA, DSA, DH, ECDH, ECDSA)	Move to hybrid cryptography, using classical and quantum protocols side-by-side	<5 years
Hash functions (SHA)	Move to larger key sizes for critical systems (e.g., SHA-512)	<5 years

Post-Quantum Cryptography (PQC) | Plan of action

- Create awareness around the quantum threat for different stakeholders
- Create an inventory of systems and applications that hold sensitive (long-lived) information
  - Map owners of and dependencies between these systems and applications
  - Inventory the cryptographic algorithms that these systems use
- Include ‘Quantum Security’ in risk assessment and procurement processes
- Create a transition plan to migrate quantum-vulnerable systems to Hybrid Cryptography
- Start migrating critical systems first, ensuring interoperability

Thank you!

Questions? ☺

Resources

1. https://learning.quantum.ibm.com/
2. https://quantum.microsoft.com/
3. https://www.feynmanlectures.caltech.edu/
4. https://www.amazon.com/Quantum-Universe-Anything-That-Happen/dp/0306821443
5. https://www.aivd.nl/documenten/publicaties/2024/12/3/het-pqc-migratie-handboek
6. https://csrc.nist.gov/projects/post-quantum-cryptography